FitnessFlex – All in one fitness platform

Group 40

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DECLARATION

I hereby declare that the work herein, now submitted as an interim report for the degree of Information Technology at Charles Darwin University, is the result of my own investigations, and all references to ideas and work of other researchers have been specifically acknowledged. I hereby certify that the work embodied in this interim report has not already been accepted in substance for any degree and is not being currently submitted in candidature for any other degree.

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LIST OF ABBREVIATIONS

AI - Artificial Intelligence

API - Application Programming Interface

HCI - Human-Computer Interaction

mHealth - Mobile Health

QR - Research Questions

O - Objective

UI - User Interface

UX - User Experience

FitnessFlex: Developing an Integrated Mobile Platform for Personalised Fitness and Nutrition Tracking

Abstract:
The rapid proliferation of smartphones has revolutionized the delivery of mobile health (mHealth) solutions, empowering individuals to take charge of their physical well-being. Despite the abundance of fitness applications, many fail to sustain user engagement due to fragmented experiences, generic content, and limited personalization. This final report presents the complete design, development, and evaluation of Fitness Flex, a cross-platform mobile application built with Flutter, aimed at integrating personalized workout tracking and nutrition management. Fitness Flex features a modern, component-based user interface, a dynamic workout system, and a robust data architecture supporting AI-powered nutrition analysis. The report details the project's background, scope, and objectives; critically reviews the literature on mHealth app success factors; describes the Agile development methodology and technical implementation; analyses user testing and performance results; and outlines future directions, including advanced machine learning integration for computer vision and conversational AI. Additionally, two proposed questions evaluates whether a unified workout nutrition flow results in task success and perceived as useful applications, and whether a validation module that provides pose-based feedback improve users confidence and efficiency during exercises sessions. The findings demonstrate that Fitness Flex addresses key gaps in the market by delivering a seamless, engaging, and personalized fitness experience.

Keywords: Mobile Application Development, Flutter, mHealth, Fitness, Nutrition, Human-Computer Interaction, Artificial Intelligence, AI Feedback

I. INTRODUCTION

Background

Mobile fitness and health applications have become an integral part of modern lifestyles, reflecting a global shift toward digital health management and self-monitoring (Curry, 2025). These applications enable users to access personalized workouts, dietary advice, and health insights conveniently from their smartphones, allowing for continuous progress tracking and goal management. The widespread adoption of such tools underscores a growing public interest in using technology to improve health outcomes and foster sustainable wellness habits.

Despite their popularity, many existing fitness applications demonstrate notable limitations in maintaining long-term user engagement and delivering individualized support. Research shows that while users often display strong initial enthusiasm, retention rates drop significantly within a few months due to repetitive content, lack of adaptability, and poor integration between workout, nutrition, and motivational components (Jacob et al., 2020). In many cases, applications rely on a “one-size-fits-all” design philosophy, providing generic recommendations that fail to accommodate differences in users’ goals, physical abilities, and behavioural patterns.

This challenge highlights the need for an intelligent and holistic approach to digital fitness solutions—one that balances functionality, customization, and technological innovation. Moreover, the absence of meaningful personalization and interactive engagement can lead to user disengagement, undermining the long-term effectiveness of digital health interventions (Chen et al., 2022). Addressing these issues requires a system capable of offering dynamic feedback, adaptive content, and seamless integration across various fitness domains.

The Fitness Flex project was conceived in response to these challenges. It aims to create a unified, user-centric platform that integrates workout tracking, nutrition management, and AI-driven exercise validation within a single application. By combining modern mobile development frameworks, intelligent analytics, and real-time feedback, Fitness Flex seeks to transform conventional fitness tracking into a personalized, engaging, and sustainable digital health experience.

Problem Statement and Significance

Fragmentation across workout logging, nutrition tracking and training guidance show significant increase in cognitive loads and often perceived as counterproductive, leading to a loss after the initial adoption (Pham et al. 2020; Monachelli et al. 2024; Lim et al. 2025). There is limited evidence reporting on whether integrating these flows alongside with lightweight, real time pose validation feedback, improve the significance in terms of task success, application usefulness and boosting user confidence during exercises. Demonstrating and evaluating such advantages would determine a practical blueprint for mHealth apps that aim to increase sustainability of user engagement in a non-clinical setting.

Research Questions and Objectives

Research Questions include:
RQ1: Does integrating workout and nutrition logging into a single flow application improve task success, time-on-task and perceived usefulness in comparison to separation flow?
RQ2: Does in set pose validation modules increase users' confidence and effectiveness during exercise?

Objectives include:
O1: Design and implementation of a cross-platform prototype that unifies all describe functions such as workout, nutrition and pose base validation.
O2: Instrument the prototype to capture usability, task outcome and AI validation accuracy.
O3: Analyses such results to indicate any bottleneck between usability and model limitations and derive a roadmap for an improved engagement and accuracy.

Project Scope

Fitness Flex is developed as a cross-platform mobile application that unifies key elements of fitness tracking, nutrition management, and AI-assisted exercise validation within a single, intuitive platform. The project aims to address common shortcomings in existing fitness apps by offering a more personalized, data-driven, and interactive experience. The initial phase of development focuses on three core functional components:

1. **Streamlined browsing interface:** enabling users to efficiently filter and explore diverse workout categories such as Strength, Cardio, and Yoga through a clean and responsive design.

2. **Interactive workout cards:** allowing users to view exercise details, mark favorites, and tailor their workout routines according to personal preferences and goals.

3. **Integrated nutrition database:** providing a foundation for users to log daily meals, monitor nutritional intake, and maintain a balanced diet in alignment with their fitness objectives.

Beyond the initial prototype, Fitness Flex expands its scope through advanced integrations that enhance usability, scalability, and engagement. These include a secure backend infrastructure using Firebase for persistent data storage, user authentication, and cloud synchronization; and AI-powered features such as a computer-vision-based nutrition scanner and the Gym Validation module, which performs real-time pose estimation, repetition counting, and corrective feedback during exercises.

Together, these developments transform Fitness Flex from a basic tracking application into a comprehensive digital fitness ecosystem capable of delivering adaptive guidance, personalized insights, and intelligent performance feedback to support users in achieving sustainable health and wellness outcomes.

Aim of Research

The primary objective of this project is to design, develop, and evaluate a fully functional prototype of Fitness Flex. This cross-platform mobile application integrates fitness tracking, nutrition management, and AI-driven exercise validation into a unified digital ecosystem. The project aims to deliver a highly engaging, user-centric experience by combining modern interface design with intelligent automation and personalized feedback mechanisms. As illustrated in Figure 1-1, the Fitness Flex ecosystem connects workouts, nutrition, and AI validation within a unified architecture.

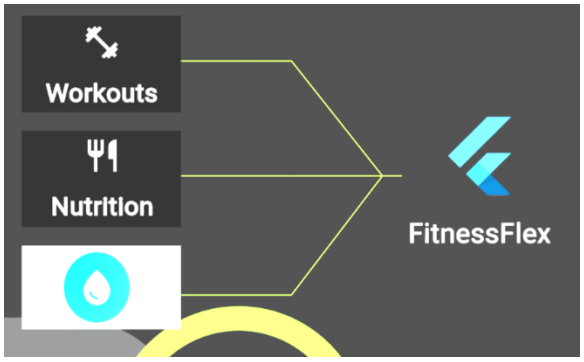

Figure 1-1. Fitness Flex Digital Ecosystem Overview.

Unlike many existing fitness applications that rely on static routines and generic interfaces, Fitness Flex leverages a component-based UI architecture, data-driven personalization, and real-time artificial intelligence to enhance interactivity and adaptability. The app incorporates computer-vision-powered workout validation through the Gym Validation module, which performs pose estimation, repetition counting, and corrective feedback to improve exercise form and safety.

By blending principles from behavioural science, user experience design, and AI technologies, the project seeks to demonstrate how intelligent mHealth solutions can promote sustained engagement, improve training effectiveness, and support long-term personal health management in an accessible and scalable manner.

Structure of Report

This final report is structured to provide a clear and logical overview of the design, development, and evaluation of the Fitness Flex mobile application:

- Section II presents a comprehensive literature review on mobile health (mHealth) technologies, highlighting user engagement strategies, personalization models, and recent advancements in AI-driven fitness applications.
- Section III outlines the methodological framework, emphasizing the Agile development lifecycle, system architecture, and cross-platform implementation using Flutter and Firebase.
- Section IV details the technical execution, including user interface development, nutrition database integration, and implementing the Gym Validation module for real-time exercise form assessment through pose estimation and repetition feedback.
- Section V presents experimental results and analysis, covering usability testing, performance evaluation, and AI validation accuracy metrics.
- Section VI discusses the system's key findings, limitations, and implications in promoting user engagement and sustainable fitness behavior.
- Section VII concludes the report by summarizing achievements and outlining future development pathways, including enhanced machine learning features and chatbot-based virtual coaching.
- The Appendices provide supplementary materials such as source code, dataset samples, test results, screenshots, and compliance documentation (e.g., similarity report, AI usage form).

II. LITERATURE REVIEW

Iintroductions of literature review

The success of mobile health (mHealth) applications depends on their ability to deliver seamless user experiences, sustain long-term engagement, and integrate emerging technologies to meet diverse fitness and wellbeing needs. Recent studies emphasize that effective mHealth solutions must balance usability, personalization, and technological innovation to promote sustained behavioural change and improved health outcomes.

This literature review examines three key dimensions that underpin the development of Fitness Flex:

1. UI/UX design principles that determine the accessibility, clarity, and intuitiveness of mHealth applications.

2. Personalization and behavioural science strategies that enhance motivation, adherence, and habit formation; and

3. Artificial intelligence (AI) integration, which enables intelligent feedback, automated nutrition tracking, and real-time exercise validation.

Together, these dimensions form the theoretical foundation for the design and functionality of Fitness Flex, guiding its evolution into an adaptive, user-centred digital health platform.

Body of literature review

1. UI/UX Design Principles in mHealth Applications:

A user-friendly interface is foundational to the adoption and continued use of mHealth applications. Nielsen (2012) emphasizes that usability defined by learnability, efficiency, and satisfaction—is critical to user acceptance and sustained engagement. In fitness contexts, this translates to streamlined navigation, clear visual hierarchies, and minimal cognitive load, allowing users to focus on achieving their health goals rather than managing complex interfaces (Schaffer & Simons, 2023).
The Mobile Application Rating Scale (MARS), developed by Stoyanov et al. (2016), identifies engagement, functionality, aesthetics, and information quality as the four core determinants of app quality. Studies indicate that applications with cluttered layouts or inconsistent workflows experience significantly higher abandonment rates (Lim et al., 2025). Consequently, adopting a component-based architecture and responsive design, as implemented in Fitness Flex, aligns with best practices for maximizing usability, scalability, and accessibility across multiple devices and user demographics.

2. Personalization and Behavioural Science for User Engagement:

While usability provides the foundation for user interaction, personalization is the key to sustaining engagement and motivation. Research suggests that tailoring content and features to individual user needs fosters a stronger emotional connection and adherence (Pham et al., 2020; Ikwunne et al., 2022). Behavioural science frameworks highlight the role of goal setting, progress tracking, and social reinforcement in enhancing user commitment to health interventions (Armitage et al., 2020).
Interactive features such as favouriting workouts, adaptive recommendations, and gamified challenges increase users' sense of ownership and accomplishment. Monachelli et al. (2024) further argue that co-creation—involving users in interface design—and adaptive interfaces that evolve with user behaviour are crucial to maintaining engagement. Similarly, gamification elements such as leaderboards, badges, and progress milestones can provide extrinsic motivation and reinforce healthy habits (Wang et al., 2024).
Fitness Flex incorporates these principles by allowing users to customize their experience through favourites and preferences. At the same time, its modular design enables the future integration of AI-driven adaptive feedback, thus blending personalization with technology-enabled motivation.

3. Artificial Intelligence in Fitness and Nutrition Applications:

The rise of artificial intelligence (AI) has revolutionized the landscape of mHealth applications, particularly in fitness monitoring, nutritional analysis, and digital coaching. Computer vision techniques have enabled automatic food recognition from images, reducing manual logging effort and improving data accuracy (Chotwanvirat et al., 2024; Li et al., 2024). Similarly, AI-powered chatbots and virtual assistants can simulate the expertise of personal trainers and nutritionists by providing real-time, context-aware guidance (Zhang et al., 2020; Kim et al., 2024).
However, despite these advancements, data privacy, algorithmic bias, and the need for explainable AI remain critical (Sicras-Mainar et al., 2023). Addressing these requires thoughtful system architecture and transparent data governance practices. Fitness Flex approaches these challenges through a modular and privacy-focused architecture that integrates its Gym Validation module—an AI-powered feature using pose estimation and repetition feedback to evaluate exercise form. This implementation exemplifies how emerging AI technologies can be harnessed to deliver personalized, safe, and data-secure digital fitness experiences.

Cconclusions from the literature review

In summary, the literature emphasizes that the success of modern mHealth applications depends on the effective integration of intuitive design, personalized engagement, and intelligent automation. A strong UI/UX foundation ensures accessibility and ease of use, while personalization strategies rooted in behavioural science foster sustained motivation and adherence. Incorporating AI technologies particularly in exercise validation, nutrition tracking, and adaptive coaching further enhances precision, interactivity, and overall user satisfaction.
These insights have directly shaped the design and development of Fitness Flex, influencing decisions across interface architecture, feature prioritization, and AI integration. Guided by established usability principles and contemporary research on user engagement, the project applies these concepts within a real-world prototype to evaluate their impact on usability and retention. The following sections detail how these theoretical foundations were operationalized through the project's methodological approach, technical implementation, and evaluation process.

III. APPROACH

The development of Fitness Flex followed a structured, yet flexible methodology grounded in Agile and Scrum principles. This approach was chosen for its adaptability, iterative structure, and emphasis on continuous evaluation qualities well-suited to mobile health (mHealth) projects where requirements often evolve throughout development (Sommerville, 2016). The project advanced through two-week sprints, with each iteration focusing on a defined set of deliverables and concluding with a supervisor review. This cycle promoted regular reflection, steady progress, and alignment with project objectives.

Team Collaboration and Supervision

The project was collaboratively developed by a two-member team, Sachin Shrestha and Sovanndevid Nong, under the supervision of Reem. Due to the small team size, responsibilities were shared dynamically across UI/UX design, frontend development, testing, and documentation.

Weekly supervision meetings with Reem were instrumental to the project's success. These sessions served as an ongoing feedback loop, where the team demonstrated progress, discussed challenges, and received guidance on technical decisions and report structure. This consistent interaction ensured that the project maintained academic integrity and professional quality.

Agile Planning and Project Management

The project adopted an Agile, sprint-based development workflow to ensure continuous progress and adaptability. Each sprint began with Sprint Planning, where user stories and acceptance criteria were defined and prioritized based on technical feasibility and value. During the Development Phase, features were implemented collaboratively, followed by a Testing and Review stage to validate functionality and fix issues. Weekly Supervisor Feedback meetings with Reem provided guidance and suggestions for improvement and ensured project alignment with academic and functional goals. Finally, the Sprint Retrospective allowed reflection on outcomes and adjustments to the backlog for the next sprint.

This iterative approach promoted flexibility, transparency, and consistent improvement throughout the project lifecycle. The process is illustrated in *Figure 1.2*, which shows how each sprint cycle feeds back into planning to support continuous enhancement of Fitness Flex.

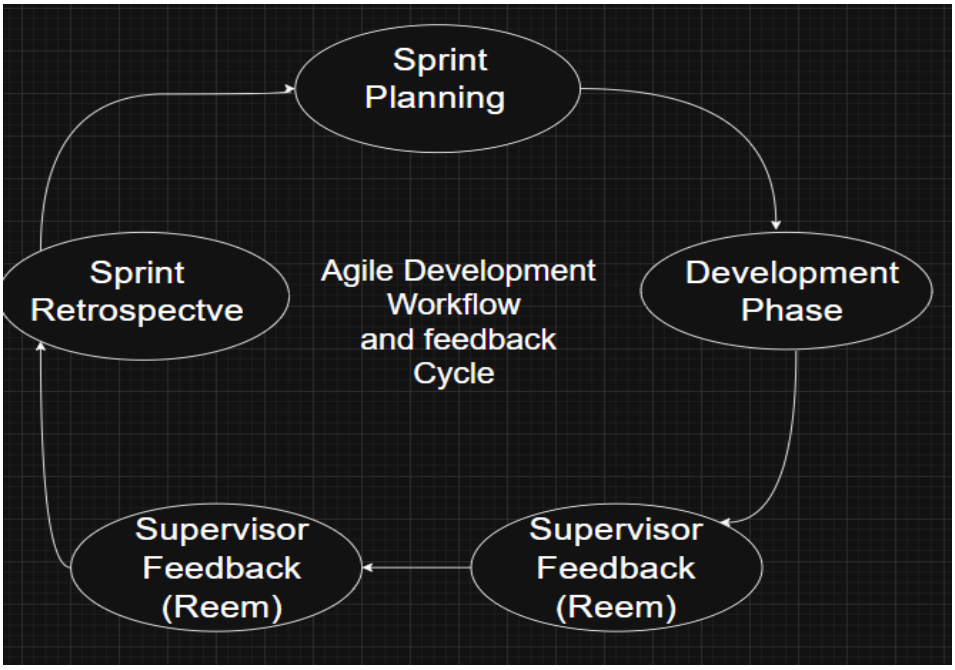

Figure 1.2. Agile Development Workflow and Feedback Cycle.

Technology Stack and Design Principles

Fitness Flex was developed using Flutter and Dart, chosen for their efficiency in building high-performance, cross-platform applications from a single codebase (Google, 2023). This approach ensures consistent performance, aesthetic coherence, and maintainability across Android and iOS devices.

Figure 1.3 illustrates that Fitness Flex's front-end architecture is structured into logical modules, including navigation management, UI screens, and reusable widgets. This modular organization enhances maintainability, scalability, and supports the integration of future functionalities. The component-based design pattern allows developers to update or extend specific modules without disrupting the entire system, streamlining long-term maintenance.

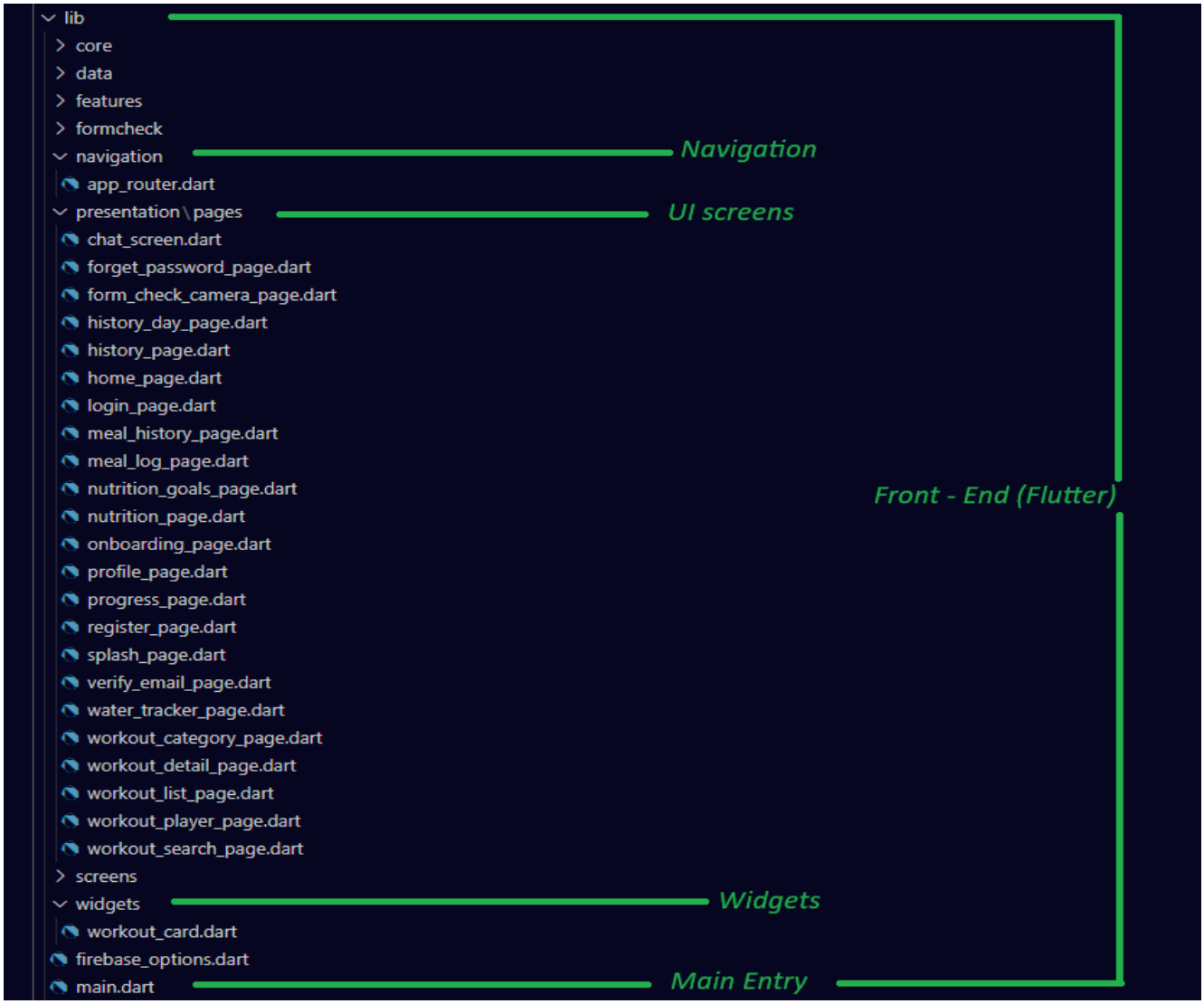

Figure 1.3 Fitness Flex Front-End Structure and System Architecture (Flutter).

Fitness Flex integrates Firebase as its backend platform to enable secure data persistence, authentication, and real-time synchronization. The backend architecture follows a clean separation between data, logic, and presentation layers. As shown in *Figure 1.4*, the application's data interaction layer is built through modular repositories such as nutrition, workout, and workout log repositories that mediate communication between the Flutter interface and Firestore. This abstraction improves code organization, reduces redundancy, and simplifies the integration of additional data models in future updates.

Figure 1.4 Fitness Flex Backend and Data Interaction Structure (Firebase Integration).

The Gym Validation Module was developed in Python as an independent AI subsystem, leveraging pose estimation and repetition detection to assess user exercise form. This module functions alongside the mobile app, processing input videos or camera feeds to analyse movement accuracy. As illustrated in *Figure 1.5*, the subsystem includes key components such as the squat validator scripts, data processing utilities (qs.py), and AI run directories for storing output logs. Together, these elements form the analytical engine that powers Fitness Flex's exercise validation feature.

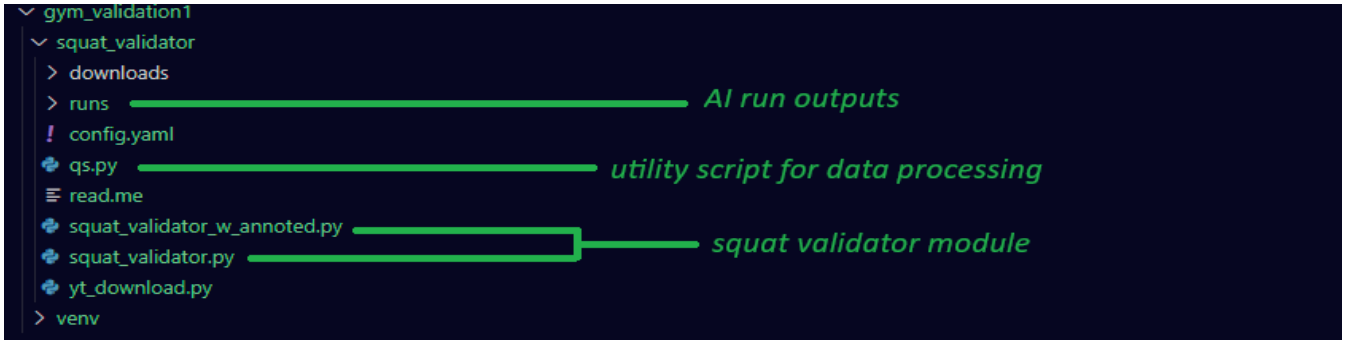

Figure 1.5 AI Module Structure for Gym Validation

As illustrated in *Figure 1.6*, the squat validation AI model produces visualized feedback by tracking body movement dynamics over time. The blue curve represents variations in knee angles, while the orange line indicates torso lean degrees. The shaded regions correspond to detected repetitions, highlighting when a complete squat is performed. These analytical outputs provide the foundation for real-time feedback and performance scoring, allowing Fitness Flex to evaluate form accuracy and consistency during exercises.

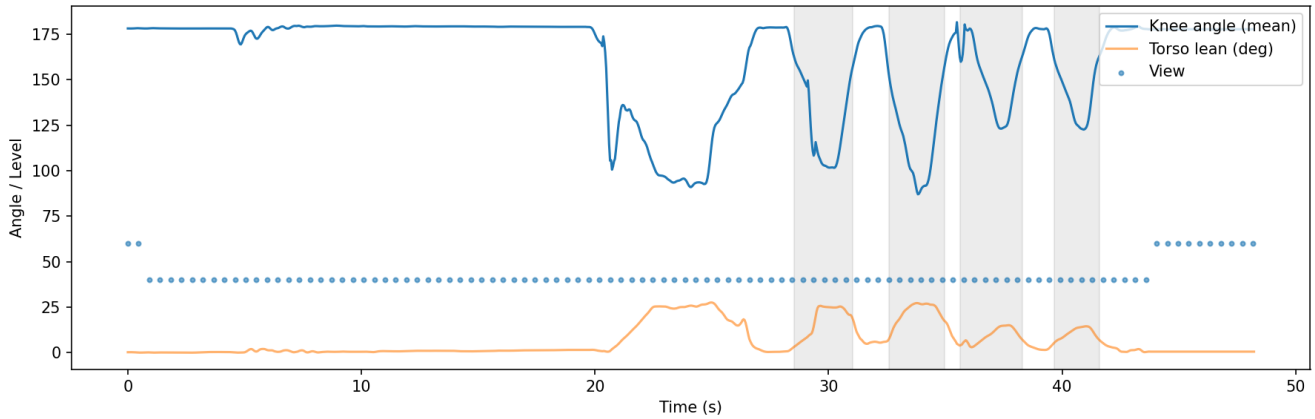

Figure 1.6 Squat Validation AI Output Showing Angle and Repetition Analysis.

Finally, a component-based design philosophy underpins the entire Fitness Flex architecture, ensuring scalability, maintainability, and reusability. Key modules such as the Workout Card, Nutrition Entry, and Gym Validation components were designed as reusable building blocks. This approach aligns with established software engineering principles (Gamma et al., 1994). It supports integrating emerging technologies such as AI-driven pose estimation, nutrition tracking, and computer vision enhancements into future releases.

Evaluation Methodology

We conducted a formative evaluation within the subject's task protocol. Participants completed(i) a unified workout-nutrition task flow and (ii) a baseline siloed flow (counterbalance). For the gym validation modules, set individuals perform guided body weight back squat while the system produced pose-based feedback.

Measurement uses such as (a) Usability: System usability scale (SUS from 0-100). (b) Task Outcome: task success rate (binary, time on task (s), and usefulness based on 7-point systems). (c) Performance: cold start time (ms), screen FPS, Firestore round-trip latency (ms). (d) Validation Accuracy: precision/recall/F1 for repetition detection against ground truth (IoU ≥ 0.5 on *start, bottom, end* windows), and timing error (mean absolute error, frames/ms) for start, bottom, and end events.

Analysis**:** We report descriptive statistics (mean $\pm$ SD) and paired comparisons where relevant; AI metrics are computed per run and macro-averaged across runs.

Quality Assurance and Version Control

Version control was managed through Git, employing feature branching, pull requests, and peer validation to maintain code quality. Manual testing and iterative debugging were conducted regularly during sprints. All changes were documented to ensure traceability and facilitate supervisor review.

The team's structured yet flexible approach enabled efficient collaboration, reflective learning, and consistent progress laying a strong foundation for the successful execution of Fitness Flex.

IV. EXECUTION

The execution phase marked the transformation of design concepts and technical plans into a functional mobile application. Development proceeded iteratively across multiple sprints, with each cycle producing a stable, testable build. Weekly meetings with Reem provided structured review opportunities, ensuring that the project remained on track and that emerging challenges were promptly addressed.

Core Application Structure

The first sprint focused on setting up the foundational framework. The team established a clear project hierarchy with distinct screens, widgets, and data models directories. This organized structure simplified collaboration and streamlined code maintenance.

A key milestone was creating the Workout Card component - a flexible, standalone widget designed to dynamically display workouts by referencing structured data. This modular approach reduced redundancy, improved scalability, and supported the rapid prototyping of user interfaces.

Workout Functionality and State Management

The Workouts Screen was implemented as a stateful widget that handled user interactions like liking and bookmarking workouts. Using Flutter's set State mechanism, the _toggleLike() function updates the UI in real time whenever users interact with workout cards. This ensured responsive and engaging feedback consistent with modern mobile app standards.

Layout Optimization and Responsiveness

During early testing, the team encountered a layout overflow error—a common issue in Flutter when content exceeds screen limits. The problem was resolved by implementing a combination of LayoutBuilder, SingleChildScrollView, and Constrained Box, ensuring adaptive layouts across multiple device types and orientations. This improvement enhanced usability and accessibility, aligning with best practices in mobile interface design (Vandad, 2021).

Nutrition Tracker Foundation

The team began developing the nutrition tracking system in parallel to the workout module. Two key data models were introduced:

- Food Entry: representing individual food items and their nutritional values.
- Daily Nutrition: aggregating daily intake data and computing calories, proteins, and carbohydrates totals.

These foundational models established the groundwork for future integration of AI-based food recognition and photo logging, supporting the long-term vision of Fitness Flex as an intelligent health management tool.

Gym Validation Module Integration

A significant advancement in the later development phase was integrating the Gym Validation module. This feature utilized pose estimation and repetition counting to evaluate users' real-time exercise form. It provided corrective feedback and recorded validation results in structured JSON files, improving accuracy and traceability. This addition transformed Fitness Flex from a conventional tracker into a semi-intelligent coaching assistant capable of promoting safer, more effective workouts.

Simulation

To run such pose-estimate-drive, rule-based state machine, the set-up sourced videos off YouTube with an estimated frame of 30fps. Each video was then process with MediaPipe Pose with the model_complexity = 1, knee angle, torso lean, and stance feature were smoothed. It then fed into a set threshold state machine to detect the start, bottom and end events per repetition performed. Ground truth events were manually annotated for better comparison.

- Configuration thresholds include: knee_standing = 155 degree, knee_bottom = 115 degree, torse_lean_ok = 10-25 degree, min_rep_seconds = 0.5, max_rep_seconds = 6.0.
- Ground Truth was done manually with result of 5 annotated repetition with start, bottom and end frames.
- Results indicated : *GT reps = 5, Pred reps = 4, Matches (IoU≥0.5) = 4 → Precision = 1.00, Recall = 0.80, F1 = 0.89*. Timing error (frames)**:** *start 10.50 ± 2.06, bottom 4.00 ± 3.08, end 17.75 ± 4.38. Converted at ~24 fps: start ~437.9 ± 86.0 ms, bottom ~166.8 ± 128.6 ms, end ~740.3 ± 182.7 ms.*

From this simulation, the rep detector indicates a conservate (no false positive) with one rep missed and accurate at the bottom with late start/end boundaries, set a solid foundation for safe feedback but with perceptible latency at rep edges.

Data Visualization and Analytics

To help users monitor progress effectively, the team implemented interactive, scrollable charts within the Progress page. These charts allowed bidirectional (horizontal and vertical) scrolling, enabling exploration of extended workout and nutrition datasets. Multiple scroll controllers and optimized rendering ensured smooth performance and usability, even when displaying large amounts of data. This design enhanced user engagement and provided a visual, data-driven feedback loop.

Testing, Review, and Continuous Improvement

Testing was performed continuously throughout development, supported by version control and sprint reviews. Both developers conducted manual testing and peer validation, ensuring functional stability before integrating new features.

Weekly supervisor meetings further guided UI, architecture, and documentation adjustments. This iterative coding, review, and refinement process reinforced the project's reliability and academic quality.

Outcome and Reflection

By the end of execution, Fitness Flex had evolved into a stable, interactive prototype featuring core fitness tracking, nutrition logging, and AI-powered gym validation. The structured workflow, consistent supervision, and strong teamwork between Sachin and Devid allowed the project to balance technical ambition with practical delivery. The outcome demonstrated the effectiveness of disciplined Agile practices, modular design, and continuous reflection in producing a user-focused, scalable mHealth solution.

V. ANALYSIS AND DISCUSSION OF RESULTS

The Fitness Flex prototype represents the culmination of iterative development, refinement, and continuous supervision. By the end of implementation, the application had evolved into a fully functional cross-platform solution that integrates workout tracking, nutrition management, and AI-assisted exercise validation within a single, cohesive platform.

User Interface and Engagement

The app opens with a clean, visually appealing Workouts Screen, designed with clarity and accessibility in mind. Users can seamlessly browse categorized workouts, explore detailed routines, and mark their favourites through the interactive "like" button. This feature employs Flutter's reactive state management, ensuring instant visual feedback and smooth performance. The design reflects sound UI/UX principles and aligns with research emphasizing the importance of personalization in sustaining user engagement (Pham et al., 2020)

Weekly review meetings with Reem contributed significantly to refining the user interface. Her feedback guided layout balance, navigation flow, and data visibility improvements, resulting in an app that feels modern, responsive, and intuitive across various device sizes.

Technical Challenges and Solutions

Early stages of development presented several technical challenges, most notably layout overflow errors caused by varying screen resolutions. These issues provided valuable learning opportunities for the team. Through the use of LayoutBuilder, SingleChildScrollView, and ConstrainedBox, the app achieved full responsiveness and visual consistency. These adaptations ensured the UI dynamically adjusted to different screens, enhancing accessibility and device compatibility. The successful resolution of these challenges reflects the team's growing technical proficiency and adherence to best practices in Flutter development (Vandad, 2021).

Nutrition Tracking and Data Architecture

A key achievement during execution was creating a robust data model underpinning the Nutrition Tracker. The Food Entry and Daily Nutrition classes were designed to capture, organize, and analyse nutritional data efficiently. This modular data structure allows for future scalability specifically, seamless integration of AI-powered food recognition and camera-based nutrition logging without major restructuring. The clear separation between data and presentation layers ensured cleaner code management and simplified future feature expansion, reinforcing the project's long-term maintainability.

Gym Validation Module Integration

A significant technical and conceptual milestone was integrating the Gym Validation module, which applies pose estimation and repetition counting to provide real-time feedback on exercise form. This feature differentiates Fitness Flex from standard fitness trackers by offering interactive guidance and performance validation, helping users perform exercises safely and effectively.

The validation process outputs structured JSON data (e.g., squat_validation.json), detailing form accuracy and detected errors—an early step toward an intelligent feedback system. While still in its initial phase, this integration demonstrates the project's readiness for future expansion into AI-driven coaching and motion analysis.

Gym Validation Module Data Analysis

The detector successfully archived perfect precision (1.00) with a recall 0.80 (F1 = 0.89), indicating that individuals are unlikely to see spurious rep counts which are important for trust and usefulness of such application. However, *start/end* binderies delay roughly between 438ms and 740ms can make the feedback feel laggy even when the rep is recognised. Two low risk tuning levers emerge:

- Sensitivity**:** slightly relax min_rep_seconds and/or decrease knee_bottom to admit **faster/shallower** reps (lift recall).
- Latency**:** add a short anticipation window for start/end transitions (or hysteresis on thresholds) to reduce boundary lag without inflating false positives.

Data Visualization and Cloud Integration

Data visualization emerged as another strong outcome of the project. The Progress page was enhanced with bidirectional scrollable charts that allow users to explore extensive workout and nutrition histories. Multiple scroll controllers and optimized rendering were used to maintain performance and ensure fluid interaction, even when handling large datasets. This feature transforms raw data into meaningful visual feedback, supporting users in monitoring their health progress over time.

In addition, the introduction of Firestore/Firebase cloud integration represented a significant advancement over the initial prototype. User preferences, workout logs, and nutrition data are now securely stored and synchronized across sessions and devices by migrating data from temporary local storage to a persistent cloud-based system. This shift strengthens the app's scalability, security, and real-world viability.

Evaluation and Reflective Insights

While large-scale user testing was outside the project's scope, internal testing and weekly supervisor reviews provided continuous evaluation and validation. These sessions emphasized usability, feature reliability, and code optimization. The feedback cycle fostered reflective learning, progressively allowing developers to refine their technical and design approaches.

The project's iterative nature grounded in Agile principles proved highly effective for a small development team. Each sprint yielded measurable progress, translating theoretical planning into tangible improvements in functionality, interface design, and overall user experience.

VI. CONCLUSIONS

The Fitness Flex project has successfully achieved its primary objective: developing a functional, scalable, and visually cohesive mobile application prototype. Using a component-based architecture, the system was designed with modularity and future extensibility much like assembling a set of reusable building blocks that can easily support additional features and enhancements.

Key functionalities, including the interactive "like" system, were implemented to promote user engagement and personalization, aligning closely with the literature emphasizing the role of adaptive design in sustaining long-term motivation (Pham et al., 2020). The application also demonstrated technical robustness by effectively handling layout responsiveness across various screen sizes, ensuring consistent usability and accessibility.

Establishing a solid data architecture for both workout and nutrition tracking was equally significant. Implementing the Food Entry and Daily Nutrition models laid a strong foundation for integrating AI-driven nutrition analysis and computer vision–based logging, positioning Fitness Flex for future development as an intelligent, data-informed health platform.

The inclusion of the Gym Validation module further extended the app's capabilities, introducing pose estimation and repetition counting to promote safe and effective workouts. This innovation reflects the project's forward-looking approach bridging mobile health with real-time performance analysis through artificial intelligence.

Beyond the technical outcomes, the project fostered collaboration, reflective practice, and continuous improvement through weekly supervisory guidance from Reem. These iterative feedback loops played a vital role in refining the application's design and documentation, ensuring academic and professional rigor throughout the process.

In conclusion, Fitness Flex has evolved from a conceptual idea into a practical, user-centred, and technically sound fitness application. The prototype provides a strong platform for future expansion particularly in integrating advanced

AI features, cloud-based analytics, and conversational interfaces. With its scalable design and commitment to personalization, Fitness Flex is well-positioned to contribute meaningfully to the growing digital health and fitness technology field.

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VII. FUTURE WORK

With the completion of the Fitness Flex prototype, the next phase of development will focus on enhancing exercise validation, expanding AI-driven nutrition analysis, and improving the system's scalability and overall user experience. The successful integration of Firestore cloud storage, robust data models, and an AI-powered chatbot has established a strong technical foundation for further advancement. The immediate priority will be to extend the Gym Validation module, which currently supports squat form analysis, to include additional exercises such as push-ups, planks, and lunges. These enhancements will leverage advanced pose estimation techniques to deliver more accurate feedback on body alignment, repetition counting, and exercise performance. Expanding this feature set will enable Fitness Flex to promote safer, more effective workouts through real-time AI-guided feedback, bridging the gap between digital coaching and physical training.

Following this, development will focus on implementing a scannable food recognition system that automatically identifies meals and estimates their nutritional content. Using computer vision and machine learning (e.g., TensorFlow Lite), the system will analyse food images to calculate calories, protein, fat, and carbohydrate levels. The data will sync with Firebase, updating users' daily nutrition summaries in real time. This feature aims to make nutrition tracking effortless, accurate, and seamlessly integrated into the overall fitness experience.

From a technical perspective, future efforts will emphasize backend optimization to improve device synchronization, authentication, and data performance. Additionally, advanced data visualization dashboards will be introduced, allowing users to monitor long-term progress and receive predictive health insights. Comprehensive usability and performance testing will ensure the app remains accessible, efficient, and engaging across different platforms and devices.

In the long term, Fitness Flex aspires to evolve into a fully integrated digital wellness ecosystem combining intelligent exercise validation, nutrition tracking, and virtual coaching. By advancing its AI capabilities and reinforcing user-centred design principles, Fitness Flex aims to redefine the digital fitness experience, transforming data-driven insights into personalized guidance for healthier and more sustainable living.

Below is the future Development plan for Fitness Flex

Phase / Feature	Description	Objective	Estimated Timeline	Expected Outcome
Phase 1: Enhanced Gym Validation Module	Extend the current squat validation feature to include push-ups, planks, and lunges using advanced pose estimation.	Improve exercise coverage and provide accurate, real-time feedback on form and performance.	4–6 weeks	Multi-exercise validation system capable of offering AI-driven corrections and repetition tracking.
Phase2: Scannable Food Recognition System	Implement image-based food logging using computer vision (TensorFlow Lite) to analyze calories, protein, fat, and carbohydrate levels.	Automate nutrition tracking to minimize manual entry and increase accuracy.	6–8 weeks	Seamless food scanning feature integrated with Firebase for real-time nutrition updates.
Phase 3: Backend Optimization and Data	Refine Firestore database performance and	Ensure secure, synchronized data storage across	3–4 weeks	Optimized backend ensuring consistent performance and

Synchronization	authentication systems to support scalability and reliability.	sessions and devices.		data integrity.
Phase 4: Advanced Data Visualization	Develop interactive dashboards and charts for tracking long-term fitness and nutrition trends.	Enhance user engagement through visually intuitive progress tracking.	3–5 weeks	Dynamic data visualization tools integrated into the Progress page.
Phase 5: Comprehensive Usability and Performance Testing	Conduct structured user testing sessions and performance assessments across devices.	Validate usability, accessibility, and engagement to inform iterative improvements.	2–3 weeks	User-tested, optimized application ready for pilot deployment and future scaling.

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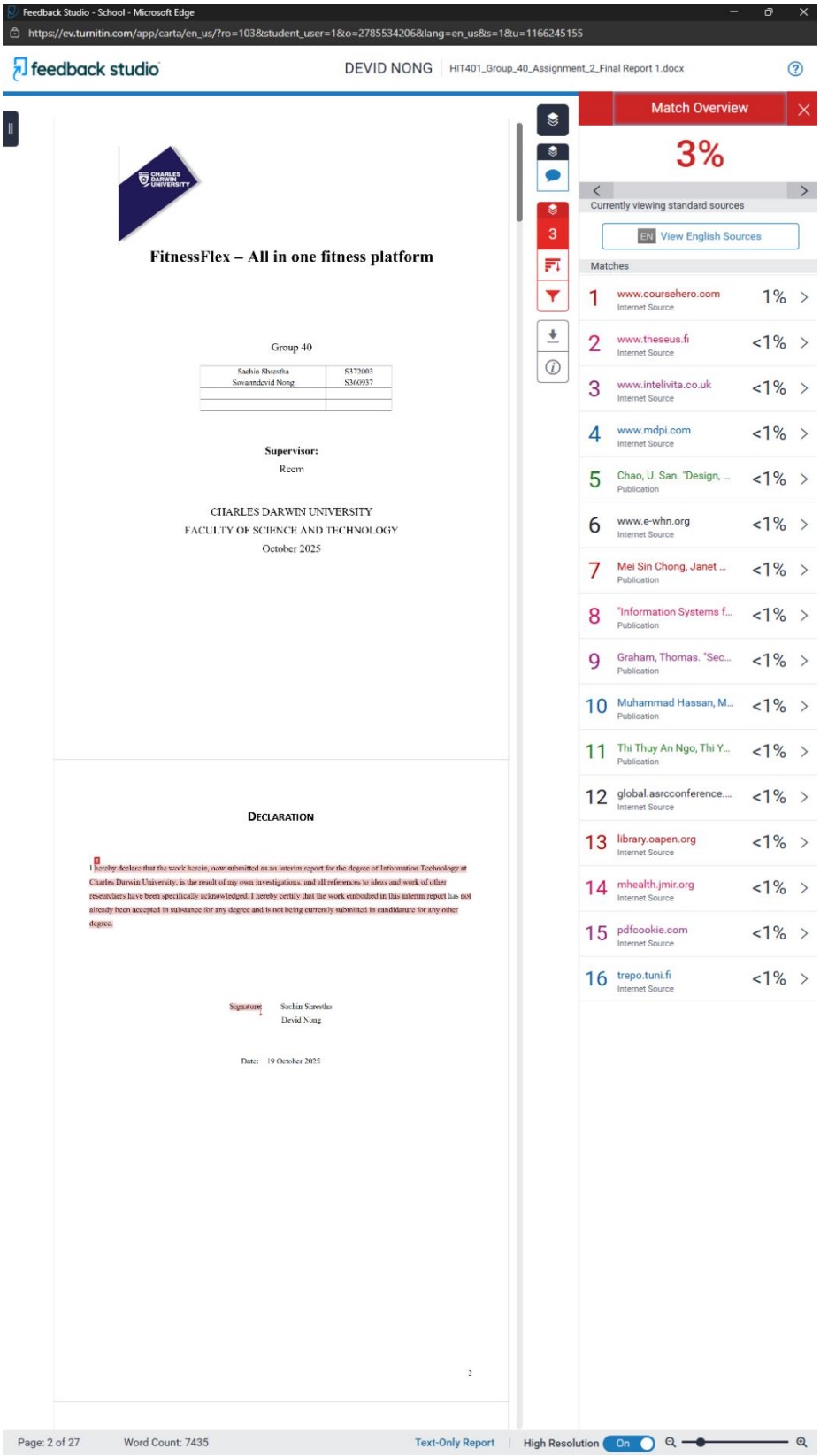
APPENDIX A: TURNITIN SIMILARITY REPORT



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FitnessFlex – All in one fitness platform

Group 40

Sachin Shrestha	S372003
Sovanndevid Nong	S360937

Supervisor:
Reem

CHARLES DARWIN UNIVERSITY
FACULTY OF SCIENCE AND TECHNOLOGY
October 2025

DECLARATION

I hereby declare that the work herein, now submitted as an interim report for the degree of Information Technology at Charles Darwin University, is the result of my own investigations, and all references to ideas and work of other researchers have been specifically acknowledged. I hereby certify that the work embodied in this interim report has not already been accepted in substance for any degree and is not being currently submitted in candidature for any other degree.

Signature: Sachin Shrestha
Devid Nong

Date: 19 October 2025

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APPENDIX B: AI DECLARATION FORM

Output text of AI tools (such as a ChatGPT 5) for written assessment items of this unit will not be treated as your own work.

- *If any output text is used, it must be quoted in italics, and the app or website must be referenced appropriately (AI tool Name, Developer or Organisation, Link).*
- *If any output ideas are used and paraphrased, the app or website must be referenced appropriately (AI tool Name, Developer or Organisation, Link).*
- *If the output of an AI tool is prescribed by your Lecturer or Supervisor as an integral part of an assessment item, the app or website must be referenced appropriately (AI tool Name, Developer or Organisation, Link).*

(excerpt from Unit Information)

Step 1: Select the declaration statement that best describes the level of AI tool usage in this assignment.

Table 1. Declaration Statement

Tick (✓) only 1 statement that applies	**Declaration statements**
	Declaration 1: We do not use any content generated by AI tools to be presented as our work.
✓	**Declaration 2:** We acknowledge the use of AI tool(s) to generate materials that aid with the process of completing this assessment (e.g. for background research or self-study). None of these materials are presented as our work.
	Declaration 3: We acknowledge the use of AI tool(s) to generate materials that are included in our work in modified forms.

Step 2: Complete the following form if you selected either Declaration 2 or Declaration 3. You do not need to complete this section if you select Declaration 1.

Table 2. Details of AI Tools Usage

Reference to one or more AI tools. Example: ***AI Tool: ChatGPT (GPT-5), OpenAI, https://chat.openai.com/***	
List all inputs(prompts) you have used.	**Outputs** (from the AI tool):
Provide a clarity, grammatical and academic tone of my draft.	Revised draft text in clearer academic tone with grammar corrections.
Suggest a credible sources for health relate mobile app.	Suggested credible academic and professional references (e.g., WHO, Bennett et al., Chen et al., Ventola).
Give me 5 top fitness app and what are their development are like.	Overview of five top fitness apps (*MyFitnessPal, Nike Training Club, Fitbit, Strava, Centr*) and their development focus (nutrition tracking, UI/UX design, device integration, social engagement, holistic wellness).
These are my assignments rubric and my draft, Act as my lecture and suggest me an estimate mark and correct on my paper.	Simulated lecturer feedback based on rubric: strengths (context, relevance, use of references), weaknesses (redundancy between sections, grammar, literature review depth), estimated mark (Credit–Distinction, 65–75%), and recommendations to reach High Distinction (expand references, refine structure, polish academic style).

Reflect on how you used these AI tools, including how they supported your assignment, the limitations you encountered, and how you modified the AI-generated output. Please provide as much detail as possible, and you may include screenshots or highlight the parts you revised.	**Response**: Utilising AI as lecture to give a rough estimated feedback which has enable us to identify our weakness in academic writing which has been refine my grammatical, suggest a credible academic sources which we verify with our own research, and taking a deeper understanding of the overall fitness sections. It has enabled us a team to understand on a deeper surface level on how it can be implemented. Although AI has been suggesting credible sources, it has been accurate enough for it to be useful and additional cross-check independently for 100% accuracy.
Do you **keep a verifiable AI prompt history accessible** to your supervisor in weekly meetings?	Yes

Student Signatures

Student ID	Full Name	Signature
S360937	**Sovanndevid NONG**	**Devid**
S372003	**Sachin Shrestha**	**Sachin**

Date:07/September/2025

The contents of this AI Usage Declaration form have been adapted from:

- University of Birmingham (2025), *Acknowledging and Citing the Use of Generative AI by Student*, accessed on 4 Mar 2025. Link: https://www.birmingham.ac.uk/libraries/education-excellence/gai/acknowledging-gai-by-students
- University of Cambridge (2025), *Template Declaration of the Use of Generative Artificial Intelligence*, accessed on 4 Mar 2025. Link: https://www.cshss.cam.ac.uk/education/generative-artificial-intelligence-ai-and-scholarship/template-declaration-use-generative

APPENDIX D: DATA AND SOURCE CODE

Main Repository : https://bitbucket.org/assesmentgroup7/group40/src/main/
Main Backend : https://bitbucket.org/devid_backend/studybuddy_backend/src
Exercises Backend : https://bitbucket.org/assesmentgroup7/exercise-backend/src/main/
Gym Validation : https://bitbucket.org/assesmentgroup7/gym_validation/src/master/

APPENDIX C: APPLICATION INTERFACE

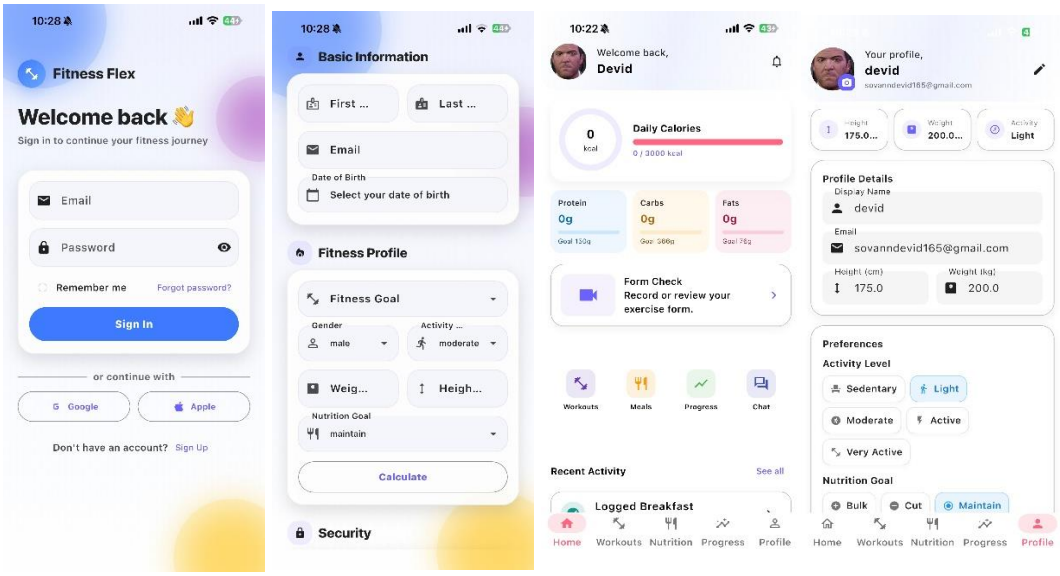

Figure 1 - User Interface

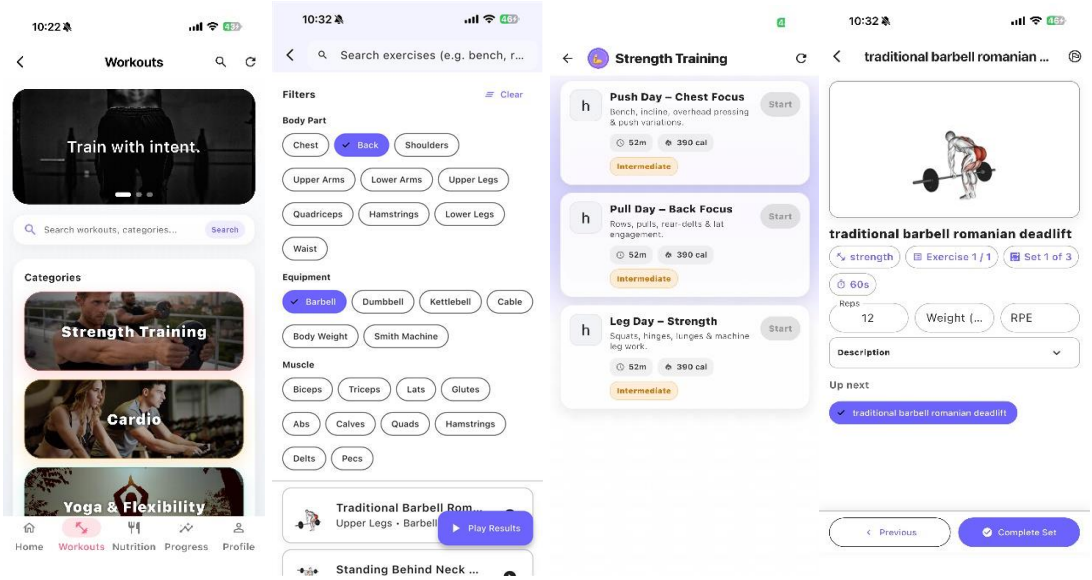

Figure 2 - Workout Interface

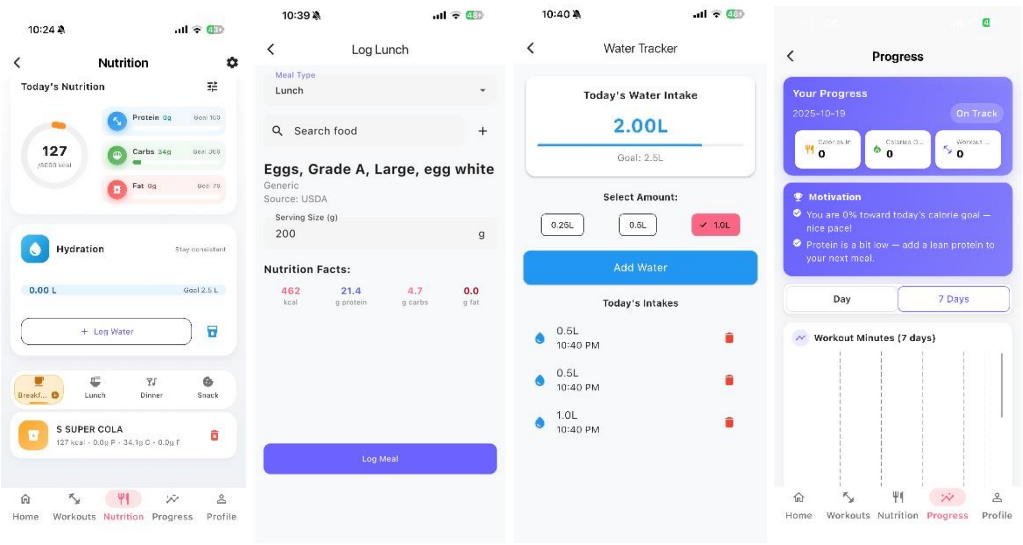

Figure 3 - Nutrition Interface

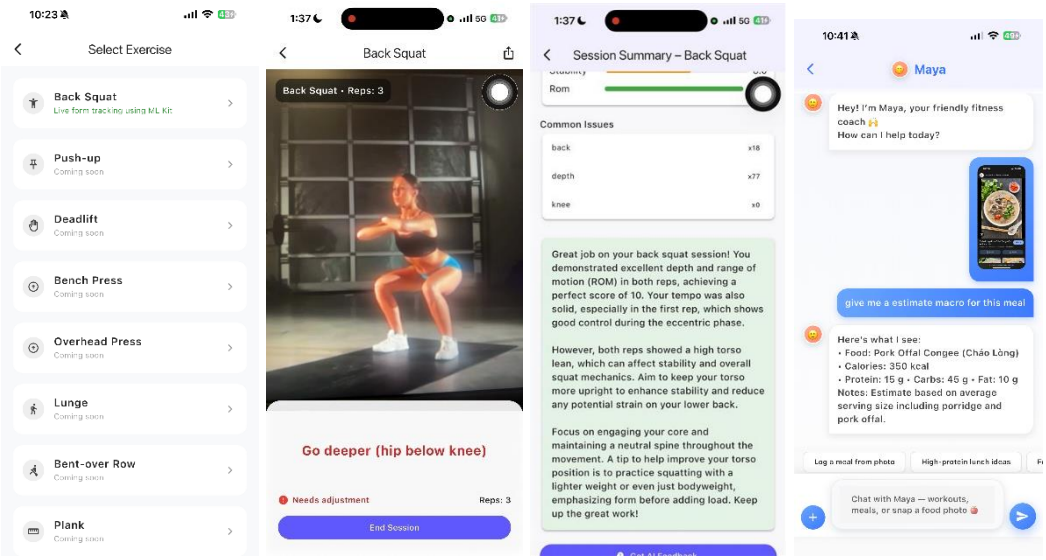

Figure 4 - Validation and Chatbot